GHULAM ISHAQ KHAN INSTITUTE
of Engineering Sciences and Technology

Faculty of Computer Science and Engineering

CE342L — Computational Methods and Tools Lab

Lab Report #3

Fixed Point Iteration Method

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Contents

Objective

The objective of this lab is to understand and implement the Fixed Point Iteration Method for solving nonlinear equations numerically using MATLAB.

By the end of this lab, students will be able to:

- Understand the mathematical formulation of Fixed Point Iteration.
- Convert nonlinear equations into the form:

$$x = g(x)$$

- Implement Fixed Point Iteration in MATLAB.
- Analyze convergence behavior for different equations.
- Visualize iterative convergence using plots.

Theory

A nonlinear equation:

$$f(x) = 0$$

can be rearranged into:

$$x = g(x)$$

Starting from an initial guess x_0, the next approximation is computed using:

$$x_{i+1} = g(x_i)$$

The process continues until:

$$|x_{i+1} - x_i| < \epsilon$$

where ϵ is the desired tolerance.

The method converges only if:

$$|g'(x)| < 1$$

near the root.

1 Task 01 — Fixed Point Iteration

Question

Solve the nonlinear equation:

$$f(x) = 2^x - 5x + 2 = 0$$

Using:

$$g(x) = \frac{2^x + 2}{5}$$

Initial guess:

$$x_0 = 0.5$$

Tolerance:

$$10^{-4}$$

Maximum iterations:

$$20$$

Theory / Manual Analysis

The equation is rearranged into:

$$x = g(x) = \frac{2^x + 2}{5}$$

Using Fixed Point Iteration:

$$x_{i+1} = \frac{2^{x_i} + 2}{5}$$

Iterations continue until the difference between consecutive values becomes less than the specified tolerance.

MATLAB Code

```
% Task 1
% f(x) = 2^x - 5x + 2 = 0

clc;
clear;

g = @(x) (2.^x + 2)/5;

x0 = 0.5;
e = 1e-4;
n = 20;

x_values = x0;

fprintf('Iteration\t x\n');

for i = 1:n

    x1 = g(x0);

```

```
    fprintf('%d\t\t %.6f\n', i, x1);

    x_values(i+1) = x1;

    if abs(x1 - x0) < e
        break;
    end

    x0 = x1;

end

fprintf('\nApproximate Root = %.6f\n', x1);

% Plot
figure;
plot(0:length(x_values)-1,x_values,'-ro','LineWidth',2);

grid on;

xlabel('Iteration');
ylabel('x Value');

title('Fixed Point Iteration - Task 1');
```

Output

C:\Users\DBZ\OneDrive\Desktop\lab3\t3.m

```
%lab3q3
g = @(x) exp(-x);

x = 0;
tol = 1e-4;
maxit = 20;
fprintf('Iter\t   x\t\t   Error\n');

for i = 1:maxit

    x1 = g(x);
    err = abs(x1 - x);

    fprintf('%d\t%.6f\t%.6e\n', i, x1, err);

    if err < tol
        break;
    end

    x = x1;
end

fprintf('\nFinal Root ≈ %.6f\n', x1);
fprintf('Total Iterations = %d\n', i);
```

Command Window

```
>> t3
Iter       x              Error
1       1.000000        1.000000e+00
2       0.367879        6.321206e-01
3       0.692201        3.243212e-01
4       0.500474        1.917271e-01
5       0.606244        1.057700e-01
6       0.545396        6.084775e-02
7       0.579612        3.421655e-02
8       0.560115        1.949687e-02
9       0.571143        1.102765e-02
10      0.564879        6.263768e-03
11      0.568429        3.549378e-03
12      0.566415        2.013992e-03
13      0.567557        1.141904e-03
14      0.566909        6.477254e-04
15      0.567276        3.673203e-04
16      0.567068        2.083338e-04
17      0.567186        1.181517e-04
18      0.567119        6.701004e-05

Final Root ≈ 0.567119
Total Iterations = 18
```

Figure 1: Task 01 — Fixed Point Iteration Convergence

Discussion

The method converged successfully and the iteration values gradually stabilized near:

$$x \approx 0.7322$$

The difference between consecutive iterations decreased steadily until convergence occurred.

2 Task 02 — Cosine Function

Question

Solve the nonlinear equation:

$$f(x) = \cos(x) - x = 0$$

Using:

$$g(x) = \cos(x)$$

with initial guess:

$$x_0 = 0$$

Theory / Manual Analysis

The equation is rearranged into:

$$x = g(x) = \cos(x)$$

Using Fixed Point Iteration:

$$x_{i+1} = \cos(x_i)$$

Since the slope magnitude near the root is less than 1, the method converges.

MATLAB Code

```
% Task 2
% f(x) = cos(x) - x = 0

clc;
clear;

g = @(x) cos(x);

x0 = 0;
e = 1e-4;
n = 20;

```

```
x_values = x0;

fprintf('Iteration\t x\n');

for i = 1:n

    x1 = g(x0);

    fprintf('%d\t\t %.6f\n', i, x1);

    x_values(i+1) = x1;

    if abs(x1 - x0) < e
        break;
    end

    x0 = x1;

end

fprintf('\nApproximate Root = %.6f\n', x1);

% Plot
figure;
plot(0:length(x_values)-1,x_values,'-bs','LineWidth',2);

grid on;

xlabel('Iteration');
ylabel('x Value');

title('Fixed Point Iteration - Task 2');
```

Output

C:\Users\DBZ\OneDrive\Desktop\lab3\t3.m

```
%lab3q3
g = @(x) exp(-x);

x = 0;
tol = 1e-4;
maxit = 20;
fprintf('Iter\t   x\t\t   Error\n');

for i = 1:maxit

    x1 = g(x);
    err = abs(x1 - x);

    fprintf('%d\t%.6f\t%.6e\n', i, x1, err);

    if err < tol
        break;
    end

    x = x1;
end

fprintf('\nFinal Root ≈ %.6f\n', x1);
fprintf('Total Iterations = %d\n', i);
```

Command Window

```
>> t3
Iter       x             Error
1       1.000000        1.000000e+00
2       0.367879        6.321206e-01
3       0.692201        3.243212e-01
4       0.500474        1.917271e-01
5       0.606244        1.057700e-01
6       0.545396        6.084775e-02
7       0.579612        3.421655e-02
8       0.560115        1.949687e-02
9       0.571143        1.102765e-02
10      0.564879        6.263768e-03
11      0.568429        3.549378e-03
12      0.566415        2.013992e-03
13      0.567557        1.141904e-03
14      0.566909        6.477254e-04
15      0.567276        3.673203e-04
16      0.567068        2.083338e-04
17      0.567186        1.181517e-04
18      0.567119        6.701004e-05

Final Root ≈ 0.567119
Total Iterations = 18
```

Figure 2: Task 02 — Convergence for $\cos(x) - x$

Discussion

The iteration values oscillated initially but gradually converged toward:

$$x \approx 0.7391$$

The convergence condition was satisfied, resulting in stable iterations.

3 Task 03 — Exponential Function

Question

Solve the nonlinear equation:

$$f(x) = e^{-x} - x = 0$$

Using:

$$g(x) = e^{-x}$$

with initial guess:

$$x_0 = 0$$

Tolerance:

$$10^{-4}$$

Theory / Manual Analysis

The equation is rearranged into:

$$x = g(x) = e^{-x}$$

Using Fixed Point Iteration:

$$x_{i+1} = e^{-x_i}$$

The iteration values gradually approach the fixed point.

MATLAB Code

```
% Task 3
% f(x) = e^(-x) - x = 0

clc;
clear;

g = @(x) exp(-x);

x0 = 0;
e = 1e-4;
n = 20;

x_values = x0;
```

```

fprintf('Iteration\t x\n');

for i = 1:n

    x1 = g(x0);

    fprintf('%d\t\t %.6f\n', i, x1);

    x_values(i+1) = x1;

    if abs(x1 - x0) < e
        break;
    end

    x0 = x1;

end

fprintf('\nApproximate Root = %.6f\n', x1);

% Plot
figure;
plot(0:length(x_values)-1,x_values,'-gd','LineWidth',2);

grid on;

xlabel('Iteration');
ylabel('x Value');

title('Fixed Point Iteration - Task 3');
```

Output

C:\Users\DBZ\OneDrive\Desktop\lab3\t3.m

```
%lab3q3
g = @(x) exp(-x);

x = 0;
tol = 1e-4;
maxit = 20;
fprintf('Iter\t   x\t\t   Error\n');

for i = 1:maxit

    x1 = g(x);
    err = abs(x1 - x);

    fprintf('%d\t%.6f\t%.6e\n', i, x1, err);

    if err < tol
        break;
    end

    x = x1;
end

fprintf('\nFinal Root ≈ %.6f\n', x1);
fprintf('Total Iterations = %d\n', i);
```

Command Window

```
>> t3
Iter        x               Error
1       1.000000        1.000000e+00
2       0.367879        6.321206e-01
3       0.692201        3.243212e-01
4       0.500474        1.917271e-01
5       0.606244        1.057700e-01
6       0.545396        6.084775e-02
7       0.579612        3.421655e-02
8       0.560115        1.949687e-02
9       0.571143        1.102765e-02
10      0.564879        6.263768e-03
11      0.568429        3.549378e-03
12      0.566415        2.013992e-03
13      0.567557        1.141904e-03
14      0.566909        6.477254e-04
15      0.567276        3.673203e-04
16      0.567068        2.083338e-04
17      0.567186        1.181517e-04
18      0.567119        6.701004e-05

Final Root ≈ 0.567119
Total Iterations = 18
```

Figure 3: Task 03 — Convergence for $e^{-x} - x$

Discussion

The method converged smoothly toward:

$$x \approx 0.5671$$

The iterations stabilized rapidly because the function satisfies the convergence condition near the root.

Conclusion

This lab demonstrated the implementation of the Fixed Point Iteration Method for solving nonlinear equations numerically using MATLAB.

- Different nonlinear equations were solved successfully using iterative techniques.
- Proper selection of the function $g(x)$ is important for convergence.
- The convergence behavior depends strongly on the derivative magnitude near the root.
- The iterative plots clearly showed how values stabilize toward the approximate solution.

Key Takeaway: Fixed Point Iteration is simple and effective when the convergence condition:

$$|g'(x)| < 1$$

is satisfied near the root.